StudyFlow: Study Planner and Focus Timer

CS310 Mobile Application Development - Project Step 1: Proposal and GitHub Setup

Application name	StudyFlow
Group number	19
Step 1 coordinator	Nazim Mammadov
Assigned TA	Zenia Mazhar
GitHub repository	https://github.com/Javidan57/studyflow

Member name	**Student ID**
Nazim Mammadov	33525
Javidan Gasimli	34580
Alimert Demirel	35700

1. App Overview

What it does. StudyFlow is a Flutter mobile app that helps students organize their courses, track assignment and exam deadlines, and study in focused sessions using a built-in timer. Everything is stored on the device, so the app works offline.

Problem addressed. University students struggle with several courses with deadlines spread across different platforms and often have no structured way to plan study time. This leads to missed deadlines and unproductive study sessions. StudyFlow keeps deadlines and study sessions in one simple place.

Intended users. University students, particularly those with multiple concurrent courses who want a lightweight, offline tool without accounts or subscriptions.

2. Core Features and Scope

Core features (planned for the semester):

- **Course management:** add, edit, and delete courses with a name and label
- **Task and deadline management:** create tasks linked to a course, with title, due date, priority, and completion status
- **Task list views:** view tasks sorted by due date, filtered by course or by status (pending/completed)
- **Focus timer:** configurable focus and break durations, start/pause/reset, optionally linked to a task
- **Reminders:** local notifications before a task deadline and when a timer session ends
- **Study session history:** every completed focus session is saved with its course and duration

Optional features:

- Weekly study statistics with charts
- Calendar view of deadlines
- Dark mode and theme settings
- Data export/backup to a file

Note: The app has no backend, no user accounts, and no network dependency, which keeps the project manageable. The core features can be completed first, and optional features are added incrementally.

3. Data Requirements

StudyFlow stores all data locally in a **SQLite database** (using the sqflite or Drift package). No external API and no cloud database are required. Simple user settings (timer durations, theme) are stored with shared_preferences.

Entity	**Stored information**
Course	id, name, label
Task	id, course id, title, notes, due date, priority, is completed
Study session	id, course id, optional task id, start time, duration (minutes), type (focus/break)
Settings	focus length, short/long break length, notifications on/off, theme

4. Potential Challenges

- **Local notifications:** scheduling reminders reliably on phones, including permission handling and time zones.
- **Background timer behavior:** keeping the timer accurate when the app is minimized or the screen is locked.
- **State management:** keeping the task list, timer, and history consistent across screens
- **Database design and migrations:** relating courses, tasks, and sessions, and updating the schema without losing data
- **Date and time handling:** sorting, filtering, and displaying deadlines correctly
- **Team collaboration:** coordinating work, making sure of no merge conflicts in a shared Git repository

5. Team Coordination and Contributions (Step 1)

Step 1 coordinator: Nazim Mammadov - organizes meetings, coordinates task assignments, and communicates with the assigned TA.

Member	**Assigned tasks for Step 1**
Nazim Mammadov	Organize meetings, communicate with the TA, final review and submission on SUCourse. Write the Core Features, Data Requirements, and Potential Challenges sections.
Javidan Gasimli	Create the GitHub repository, add collaborators (all members, instructor, and TA), add Flutter .gitignore and make the initial commit.
Alimert Demirel	Write the README.md (title, description, group details) and the App Overview section.

All members will review the final submission and be able to explain their own work.

6. GitHub Repository

- Public repository link: https://github.com/Javidan57/studyflow
- All group members are added as collaborators.
- README.md includes the project title, a brief description, and group details.
- A Flutter .gitignore file as an initial commit is present.